



Worldwide reach of short selling regulations[☆]

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ABSTRACT

We characterize legality and incidence of short selling in a worldwide, multimarket framework. Home country short selling restrictions curtail home market stock borrowing by 45% and reduce short selling of the country's American Depository Receipts (ADRs) by 68% due to regulatory reach. Also, the 2008 US ban on short selling of financial firms reduced borrowing in foreign locations. These findings are robust to controls for option availability, enforcement, returns, firm size, trading volume, dividends, ADR level, volatility, days-to-cover, and industry sector. Further, we show that investor conduct resulting from adherence to professional standards is a more powerful mechanism of regulatory reach than intergovernment cooperation.

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1. Introduction

Short selling has long been a controversial trading strategy. The academic evidence on the value of short selling is mixed, and market regulators' actions suggest a degree of ambivalence toward the practice.¹ On the one hand, regulators often publicly espouse the benefits of short selling in terms of its importance to the efficient processing of information in asset markets.² On the other

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¹ Most academic papers argue that short selling is an essential part of the price discovery mechanism (Nilsson, 2008; Boehmer, Jones, and Zhang, 2009; Kolasinski, Reed, and Thornock, in press), but there are exceptions (Shkilkov, Van Ness, and Van Ness, 2012). Nonetheless, Diether, Lee, and Werner (2009) find that the practice is pervasive and short selling volume is 24% of NYSE and 31% of Nasdaq trading volume. Edwards and Hanley (2010) find that short selling begins right from the day when a stock is sold in the initial public offering.

² For example, the Financial Services Authority (www.fsa.gov.uk/pubs/discussion/dp17_newsletter.pdf), the Nasdaq (<http://www.nasdaq.com/quotes/short-interest.aspx>), Australian Securities and Investments Commission chairman Greg Medcraft (www.theaustralian.com.au/busi

hand, these same regulators typically react to periods of market turmoil by banning short selling, arguing that the practice exacerbates market volatility and, in the extreme, destabilizes markets. The 2008 financial crisis is no exception, and many regulators either resorted to outright bans on short selling or imposed trading restrictions in an effort to stem the falling market and reduce volatility. Bris, Goetzmann, and Zhu (2007), Charoenruek and Daouk (2009), and Beber and Pagano (2013) all provide evidence on the cross-country and time series variation in short selling regulations.

The complexity of this issue is further exacerbated by the fact that, while the motivation for the bans is often clear, whether or not the bans were in any way successful in achieving these goals is entirely unclear. The limited empirical research suggests that the bans did not reduce volatility (Boulton and Braga-Alves, 2010; Saffi and Sigurdsson, 2011) and could even have increased volatility (Boehmer, Jones, and Zhang, 2009; Charoenruek and Daouk, 2009). Securities and Exchange Commission (SEC) chairman Christopher Cox and commissioner Kathleen Casey expressed the view that 2008 short selling bans created significant disruption and distortions in the market (www.washingtonpost.com/wp-dyn/content/article/2008/12/23/AR2008122302765.html and www.theage.com.au/business/us-regulator-short-selling-ban-was-disruptive-20090303-8ne6.html). Other regulators do not appear to share this view, however, and short selling bans were again imposed in many countries in 2011.

The purpose of our study is to provide further evidence on the effectiveness of short selling restrictions. However, unlike previous research, our paper is the first to investigate how home market restrictions affect short selling in a global multi-market setting. Specifically, we examine the extent to which national regulators are able to effectively enforce short selling restrictions both within and outside their home markets. Our focus is on firms that are cross-listed in the form of an American Depositary Receipt (ADR), as they provide an ideal setting for testing our research questions.

In our paper, two competing hypotheses are considered to explain the impact of short selling restrictions on foreign markets—regulatory arbitrage versus *regulatory reach*. The regulatory arbitrage hypothesis suggests that when the home market introduces restrictions, short selling moves to foreign locations. In the current context, this suggests that regulatory arbitrage increases ADR short volume if traders opt to trade in unrestricted regimes, although the existence of taxes or fees on foreign transactions, capital controls, inconvertibility of currencies, and market segmentation does complicate the relation (Foerster and Karolyi, 1999). Empirical support for regulatory arbitrage hypothesis can be found in Blau, Van

Ness, and Warr (2012), who construct a feasibility index of short selling based on subjective assessment of survey responses and find evidence of increased shorting in the United States of ADRs from countries in which short selling is arguably not feasible in the domestic market. An important point of distinction between Blau, Van Ness, and Warr (2012) and our paper is that we have invested substantial effort in ensuring that our legality of short selling variable accurately reflects the status of short selling regime in each country over time.³ Moreover, our comprehensive data sample covers 1,035 ADRs from November 2007 to December 2010 compared with only 352 ADRs sampled over 2005 and 2006 in Blau, Van Ness, and Warr (2012).

The regulatory reach hypothesis suggests that home country restrictions curtail short selling of cross-listed stocks in foreign markets. In the current context, this suggests that foreign country trading restrictions decrease short selling of a stock's ADR in the US market. Regulatory reach can decrease short selling of a country's ADRs in the United States through a variety of mechanisms that we group into two major themes: intergovernment cooperation and investor conduct. Regulatory reach could reflect a country's bilateral investment treaties or membership in groups such as G7, the Organization of Economic and Co-operation and Development (OECD), or the European Union (EU) that facilitate intergovernment cooperation through clearer communication of the regulatory intent to foreign market participants (Lau and McNish, 2002), the court's recognition and enforcement of foreign laws (Keller, 2004), and cooperation among global law enforcement agencies (Block, 2007; Hamilton, 2008). For example, in 2008 the UK's Financial Services Authority (FSA) issued Short Selling (No. 3) Instrument 2008/51, which, according to Avgouleas (2010, p. 17), "had a global reach covering shorting of shares in the list anywhere, e.g., on Frankfurt (Deutsche Borse) or the New York Stock Exchange." Similarly, in its statement concerning short selling, the SEC said that its actions were taken in consultation with regulators of the major developed securities markets around the world with whom it coordinated in monitoring market reactions (<http://www.sec.gov/news/press/2008/2008-235.htm>).⁴ The International

(footnote continued)

[ness/markets/securities-regulator-will-not-ban-short-selling-says-greg-medcraft/story-e6frg916-1226113803879](http://markets/securities-regulator-will-not-ban-short-selling-says-greg-medcraft/story-e6frg916-1226113803879)), and Securities and Exchange Commission commissioner Kathleen Casey (<http://www.sec.gov/news/speech/2010/spch022410klc-shortsales.htm>) have all stated that short selling is a legitimate investment activity, which plays an important role in supporting efficient markets.

³ Our effort corrects some erroneous classifications adopted in the prior research. For example, we are claiming that short selling in Spanish ADRs is high because it is legal to short sell in Spain (regulatory reach). Blau, Van Ness, and Warr (2012) conclude that short selling in Spanish ADRs is high because short selling in Spain is not feasible (regulatory arbitrage). The position taken in prior research is incorrect because our stock borrowing data show that it is very feasible to short sell in Spain. This observation is true for several countries where we observe short selling-related borrowing (which has a correlation of 0.9 with short interest according to Data Explorers) occurring even though short selling is classified as infeasible.

⁴ Similarly, Rodrigo Buenaventura from the Comisión Nacional del Mercado de Valores commented that one of the reasons short selling was banned in Spain during the 2008 crisis was to avoid attracting the pent-up short selling demand that could not be expressed in other euro markets, which had bans in place ("Short Sales Restrictions—What Are They Good For?" panel discussion at the IX Madrid Finance Workshop (Short Selling), Instituto de Estudios Superiores de la Empresa, Madrid, Spain; November 4, 2010). Also, on October 5 Authority for the Financial

Organization of Securities Commissions (IOSCO) has put in place a structure for its member agencies to cooperate and exchange information in the process of developing, implementing, and promoting adherence to internationally recognized and consistent standards of regulation, oversight, and enforcement.

Alternatively, regulatory reach could be driven by investor conduct and worldwide compliance. For example, financial firms could avoid circumventing regulations in their home markets to avoid scrutiny by the regulator and possibly violating the terms of their trading license (Ellickson, 1991). Pagano, Panetta, and Zingales (1998), Coffee (1999, 2002), and Erickson, Goolsbee, and Maydew (2003) observe that the overwhelming majority of firms take a conservative stance and do not engage in regulatory arbitrage at all. Further, many individuals employed within the finance industry are governed by the CFA Institute's Code of Ethics and Standards of Professional Conduct, which states that members must understand and comply with all applicable laws, rules, and regulations of any government, regulatory organization, licensing agency, or professional association governing their professional activities. In the event of conflict, members must comply with the more strict law, rule, or regulation. Similarly, the CFA Institute's standard relating to integrity of capital markets and market manipulation could discourage members from undertaking stock lending and other closely related activities when short selling is unlawful.

Our paper assembles a unique panel database from a variety of sources that allows us to establish the relative effectiveness of short selling restrictions, including the relative merits of the regulatory reach versus regulatory arbitrage hypotheses. The distinction between these two hypotheses is important because, in a race to the bottom (Schram, 2000), regulatory arbitrage implies that stricter rules (whether excessive or fair) decrease a country's business competitiveness relative to other countries with lax rules. Therefore, regulatory reach is essential to obtain the desired investor protection outcomes with reasonable costs in a competitive global economy.

The main results of our paper can be summarized as follows. We begin by considering the effectiveness of regulatory enforcement of short selling restrictions in the home markets. While intuitively obvious, empirical work on this issue is lacking, and, as expected, we find that short selling-related stock borrowing is significantly lower in countries that impose restrictions. To understand the dynamics of a multi-market trading environment, several unique short selling data sets containing stock borrowing, short interest, and short trading volume are combined. Univariate comparisons, matched control sample experiments, and multivariate regressions all consistently provide evidence in support of the regulatory reach hypothesis. That is to say, restrictions curtail short selling not only in the home market, but also in the US market

where the securities are cross-listed as ADRs. These findings remain strongly significant after controlling for option market availability, firm-specific characteristics such as past returns, return volatility, size, dividend policy, ADR level, stock borrowing costs, industrial sectors, and trading volume. Furthermore, regulatory reach works for the US firms in the reverse direction. Specifically, in September 2008, when the United States banned short selling of financial firms, stock borrowing for those firms declined in the foreign locations outside the United States as well. Finally, our tests of the mechanisms of regulatory reach indicate that investor conduct is a stronger mechanism of regulatory reach than intergovernment cooperation.

2. Data sources and sample formation

Firms with cross-listed ADRs provide an ideal setting for testing our research questions. As such, an initial sample of 2,892 eligible firms was obtained, after removing the firms that are listed on exchanges outside the United States and firms that do not have information on International Securities Identification Number (ISIN), from the list of 3,013 ADRs downloaded in November 2009 from the Bank of New York Mellon and J.P. Morgan websites www.adrbnymellon.com/dr_directory.jsp and www.adr.com/DRSearch/CustomDRSearch. Each firm's home country is identified using the first two digits of the ISIN that represents the originating country's ISO (International Organization for Standardization) codes. Information on firm-specific returns, market capitalization, volume, dividend yield, closing price, intraday high price, intraday low price, and sector classification is sourced from Datastream International.

For each of these ADR-issuing firms, data on short selling are assembled from a variety of sources. Daily information on the stock lending for the period July 2006 to January 2010 is obtained from Data Explorers. Data Explorers report on their website www.dataexplorers.com that they cover 70% of worldwide stock borrowing and that the correlation between publicly reported level of short interest and the level of stock lending is approximately 0.90. These statistics suggests that stock lending data provide a reasonable proxy for short selling. In addition to individual stock loan information, Data Explorers provides information on the amount of stock available for loan and the stock borrowing cost.⁵ The database allows the identification of the country in which the borrowing takes place, so it is possible to tell whether the shares are being borrowed in the home country or elsewhere. A more detailed description of the Data Explorers data set can be found in Saffi and Sigurdsson (2011).⁶ After merging the initial sample of ADRs with

(footnote continued)

Markets in Netherlands amended the restriction on short selling to reflect the UK FSA's restriction and disclosure requirement (2009 client briefing by Clifford Chance LLP.

⁵ Stock borrower indirectly pays the lending fee to the stock lender. Basically, the lending fee is an imputed cost. It is the difference between regular interest rate and the rebated interest (rebate rate) paid by the security lender to the security borrower on the cash collateral that the borrower provides to the lender.

⁶ Data Explorers data are cleaned as follows. We eliminate exact duplicates. Data Explorers provides values in one of four currencies—USD

Data Explorers stock borrowing data and Datastream firm characteristics data, our final daily data sample has 1,601 ADRs to assess regulatory effectiveness.

In addition to the daily Data Explorers stock borrowing information, we use fortnightly short open interest data from shortsqueeze.com for more than 16 thousand stocks that trade on NYSE, Nasdaq, Amex, OTC Bulletin Board (OTCBB), and Pink Sheets. Short interest is defined as the total number of outstanding shorted shares for each security. In addition to the number of shares sold short, supporting data such as days to cover and institutional ownership are available. Triangulation of the initial ADR list, the shortsqueeze.com open interest data, and Datastream yields a fortnightly data sample of 1,035 ADRs.

Our final sources of short selling data are Financial Industry Regulatory Authority (FINRA, Regulation-SHO data) and the BATS Global Markets Exchange. These transaction-level short sale data are aggregated to obtain daily short volume for each stock from the first availability date of August 2009 to January 2010 (the latest period for which we have Data Explorers data). Triangulation of the initial ADR list, the aggregated daily FINRA and BATS data, and Datastream yields a trading data sample of 556 ADRs.

3. Short selling regulations around the world

Information on the legality of short selling across a wide range of countries is obtained by cross-referencing the survey articles of Bris, Goetzmann, and Zhu (2007), Charoenrook and Daouk (2009), and Beber and Pagano (2013) with the practitioner's report such as 2009 client briefing from Clifford Chance LLP website http://www.cliffordchance.com/publicationviews/publications/2009/07/clifford_chance_commentsshortsellingrules.html.

Where clarification or further information is necessary, we directly correspond with the stock exchanges and financial market regulators. Table 1 presents a summary of information on the legality status of short selling in each country with specific details of the periods when short selling was legal or illegal. In addition to considering a simple binomial measure of short selling legality, we gain insights into the impact of short selling restrictions by distinguishing between the different short selling rules. The details of any intermediate forms of restrictions are presented in Columns 1–3. Apart from outright bans, other forms of restrictions include specific trading mechanisms (uptick rule), pre-borrowing requirements (ban on naked short selling), and bans on shorting selected stocks (typically financial stocks).

(footnote continued)

(US dollar), EUR (euro), JPY (Japanese yen), AUD (Australian dollar)—depending on where the data are collected. We convert all monetary values into USD, based on daily exchange rates obtained from Datastream. Data Explorers includes the variable *dividend requirement* that allows us to make sure that our results are not affected by dividend capture trades. Only the observations with *dividend requirement*=100 are kept because these are the standard loan agreements. *Dividend requirement* other than 100% indicates dividend capture and tax arbitrage-motivated trades.

Similar to previous studies, we create an indicator variable, *illegal*, which equals one if short selling is prohibited in the home market and zero otherwise. Unlike past research, however, which typically specifies an unconditional measure of short selling restrictions, our research design allows the *illegal* variable to vary over time for countries that changed their short selling rules during the sample period. For example, 19 countries imposed temporary restrictions on short selling during the 2008 financial crisis, while China lifted its restriction on short selling for 11 brokerage firms in 2008. The intermediate forms of restrictions create interesting cross-sectional variations for our analysis.

Table 1 also presents basic metrics on the relative presence of short selling in each country. Mean and median of the aggregate dollar amount of short selling-related borrowing of all stocks from each country are reported in Columns 4 and 5. Mean and median scaled borrowing ratio (SBR) for each country, which is calculated as the daily average outstanding dollar borrowing divided by the country's total stock market capitalization at the end of previous year, are reported in Columns 6 and 7. Finally, mean and median of the ADR short interest ratio, defined as the short interest as a percentage of shares float, are reported in Columns 8 and 9.

Preliminary insights into the impact of regulations on short selling can be obtained by considering a plot of short selling in home markets as well as in the ADR markets. Sample firms are divided into unrestricted and restricted categories based on home-country short-selling regulatory regimes. The restricted category includes uptick restriction, naked ban, or a total ban on short selling. Fig. 1, Panel A plots the average home market stock borrowing in millions of US dollars (USD) for the countries in the two groups. In Fig. 1, Panel B stock borrowing is scaled by the firm's market capitalization. Both panels indicate that short selling restrictions are effective in curtailing home market stock borrowing because the stock borrowing for unrestricted countries is higher than the stock borrowing for restricted countries.

Next, short interest and short volume data are used to test the regulatory reach hypothesis as opposed to the regulatory arbitrage hypothesis. Fig. 2, Panel A plots the short interest of ADRs originating from the countries in which short selling is legal and the short interest of ADRs from the countries in which short selling is illegal. The short interest of ADRs from the countries in which short selling is legal is generally higher (44 out of 47 fortnights) than the short interest of ADRs from the countries in which it is illegal. Fig. 2, Panel B presents a similar plot for short volume and shows that short volume of ADRs from the countries in which short selling is legal is always higher than the short volume of ADRs from the countries in which it is illegal. Thus, these results provide initial evidence in support of the regulatory reach hypothesis.

4. Empirical results

This section begins with a brief description of our general framework for the regression models and then

Table 1

Global shifts in short selling regulations and activity.

This table provides a timeline of the legality of short selling around the world, including the dates of bans implemented during the financial crisis of 2008. Details of both borrowing in the home country and short selling of American Depository Receipts (ADRs) in the United States are included. Column 1 (Column 2) has periods when short selling was legal (illegal) in a particular country. For these two columns, the periods are classified as illegal based on a total ban. None, always, and since inception refer to periods within our sample period. Column 3 has other short selling restrictions when short selling is not completely banned. Columns 4 and 5 have mean and median daily average outstanding dollar value of shares borrowed summed across all stocks from that country. Columns 6 and 7 have mean and median scaled borrowing ratio (SBR), which is the daily average outstanding dollar borrowing during our sample period (July 2006–January 2010) divided by the country's total stock market capitalization at the end of the previous year. Columns 8 and 9 have mean and median ADR short interest ratio, which is defined as the short interest as a percentage of share float. ^aIndicates that stock borrowing information is unavailable or missing in Data Explorers data. ^bIndicates that the United States does not have any ADRs for domestic companies. Canadian stocks also are fully fungible and do not trade as ADRs. Hence, the United States and Canada are excluded from the remainder of our analysis.

Country	Period when legal	Period when illegal	Nature of restriction and other comments	Total borrowing (millions of dollars)		SBR		ADR short interest ratio	
				Mean	Median	Mean	Median	Mean	Median
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Argentina ^a	Since 1999	Before 1999	Uptick rule applies; naked short selling prohibited	0	0	0	0	0.86	0.13
Australia	Pre-September 22, 2008; November 20, 2008–present	September 22, 2008–November 19, 2008	Naked short selling prohibited since 2001; Ban on shorting financial stocks, September 22, 2008–May 25, 2009	30,258	27,441	2.93	3.13	0.35	0.11
Austria	Since inception	None	Ban on naked short selling of financial stocks, October 27, 2008–November 30, 2010	3,777	3,030	2.55	2.29	0	0
Bahrain ^a	None	Always		0	0	0	0	0	0
Bangladesh ^a	None	Always		0	0	0	0	0	0
Barbados	None	Always		0.01	0	0	0	No ADRs	
Belgium	Since inception	None	Ban on naked short selling of financial stocks, September 22, 2008–September 21, 2009	6,720	6,587	2.07	1.88	0.52	0.10
Bermuda	None	Always		8,985	6,816	0	0	0	0
Brazil	Since inception	None	Naked short selling prohibited	22	14	0	0	10.03	1.13
Bulgaria ^a	None	Always		0	0	0	0	0	0
Canada ^b	Since inception	None	Ban on shorting financial stocks (including interlisted in US), September 19, 2008–October 8, 2008; uptick rule applies	58,183	55,440	3.75	3.71	No ADRs	
Cayman Islands	Since inception	None	Very little trading occurs on the stock exchange	4,395	4,058	0	0	6.71	6.38
Chile ^a	Since 1999	Before 1999	Uptick rule applies; naked short selling prohibited	0	0	0	0	0.36	0.18
China	None	Always	In September 2008, China allowed short selling of 11 brokerage firms	9,440	8,255	0.35	0.29	0.87	0.30
Colombia ^a	None	Always		0	0	0	0	0.74	0.48
Croatia ^a	None	Always		0	0	0	0	0	0
Cyprus	None	Always		173	139	1.14	0.89	0	0
Czech Republic	Since inception	None		104	76	0.18	0.14	0	0
Denmark	Since inception	None	Ban on shorting bank stocks, October 13, 2008–present	3,650	3,394	1.77	1.61	0.13	0.13
Ecuador ^a	None	Always		0	0	0	0	0	0
Egypt ^a	None	Always		0	0	0	0	0	0
Finland	Since 1998	Before 1998		6,423	5,625	2.62	2.51	0.55	0.52
France	Since inception	None	Ban on naked short selling of credit institutions and insurance companies' stocks, September 22, 2008–present	102,719	91,787	4.80	4.51	2.63	0.10
Georgia ^a	None	Always		0	0	0	0	0	0
Germany	Since inception	None	Investment funds except hedge funds cannot short sell; ban on naked short selling of specified financial stocks, September 19, 2008–present	81,259	67,441	5.11	4.42	0.53	0.17
Greece			Uptick rule applicable before 2008 and after May 2009; naked short selling prohibited	177	142	0.12	0.09	0.08	0.04

Table 1 (continued)

Country	Period when legal	Period when illegal	Nature of restriction and other comments	Total borrowing (millions of dollars)		SBR		ADR short interest ratio	
				Mean	Median	Mean	Median	Mean	Median
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Hong Kong	Pre-October 10, 2008 ; June 1, 2009–present Since 1994	October 10, 2008– May 31, 2009 Before 1994	Permitted for specified securities (33 in 1994–1995); uptick rule applies; naked short selling prohibited	7,561	6,774	0.01	0	6.29	1.80
Hungary	Since 1996	Before 1996		905	900	2.89	2.58	0.05	0.04
Iceland	Since inception	None	Ban on naked short selling of financial stocks, November 6, 2008–January 31 and 2009	47	15	0.14	0.05	No ADRs	
India	Since December 20, 2007	Before December 20, 2007 (Badla trading existed)	Badla trading–carry over transaction with extended rolling settlements; naked short selling is prohibited; on October 20, 2008, Securities and Exchange Board of India disapproved stock lending by foreign institutional investors of participatory notes stocks	1	0	0	0	1.26	0.91
Indonesia	Pre-October 2008; May 2009–present	October 2008–April 2009	Legal only for specified stocks	29	28	0.02	0.02	0.24	0.22
Ireland	Pre-September 19, 2008	September 19, 2008–present		2,349	1,729	2.10	2.23	0.97	0.16
Israel	Since inception	None	Naked short selling prohibited	293	257	0.18	0.16	1.09	0.52
Italy	Since inception	None	Financial stocks: naked ban, September 23, 2008–December 29, 2008; total ban from December 30–31 may 2009; naked ban June 1, 2009 –July 31, 2009, Non financials: naked ban October 10, 2008–July 31, 2009	29,328	26,727	3.41	3.08	1.66	0.24
Jamaica ^a	None	Always		0	0	0	0	0	0
Japan	Since inception	None	Uptick rule and locate requirement apply; ban on naked short selling, October 30, 2008–present	47,580	50,822	1.12	1.14	0.30	0.04
Jordan ^a	None	Always		0	0	0	0	0	0
Kazakhstan ^a	None	Always		0	0	0	0	0	0
Kuwait ^a	None	Always		0	0	0	0	0	0
Latvia ^a	None	Always		0	0	0	0	0	0
Lebanon ^a	None	Always		0	0	0	0	0	0
Lithuania ^a	None	Always		0	0	0	0	0	0
Luxembourg	Since inception	None	Ban on naked short selling of banks and insurance companies, September 19, 2008–present	5,100	5,044	5.98	6.09	3.19	1.45
Malaysia	Pre-1997; January 2007–present	September 1997–December 2006	Naked short selling prohibited; uptick rule applies; legal only for specified stocks	5	3	0	0	0	0
Malta ^a	None	Always		0	0	0	0	0	0
Mauritius	None	Always		23	12	0.59	0.28	0	0
Mexico	Since inception	None	Naked short selling prohibited; uptick rule applies	1,043	1,000	0.33	0.35	1.33	0.41
Morocco	None	Always		35	35	0.08	0.05	0	0
Netherlands	Since inception	None	Naked short selling ban, September 22, 2008–June 1, 2009; Additional restrictions based on UK FSA's requirements, October 2008	18,453	17,269	2.79	2.65	0.92	0.47
New Zealand	Since 1992	Before 1992	Since April 1992, specified securities eligible for short selling; after July 2000, all liquid securities eligible; short selling is hindered by tax legislation	760	809	1.72	1.84	0.09	0.09
Nigeria	None	Always		0.10	0.03	0	0	0	0
Norway	Since 1992	None	Ban on naked short selling of five specified financial stocks, October 8, 2008–present	5,803	4,280	2.33	1.89	0.32	0.16
Oman ^a	None	Always		0	0	0	0	0	0
Pakistan ^a	Since inception	None	"Regulations for Short Selling under Ready Market" adopted in 2002 banned naked short selling, allowed short selling only in prescribed securities; uptick rule applies	0	0	0	0	0	0

Panama	None	Always		959	1,069	16.44	18.92	0	0
Peru ^a	None	Always		0	0	0	0	0.96	0.74
Philippines	Since 1998	Before 1998	Naked short selling prohibited; uptick rule applies; legal only for specified stocks	6	6	0.01	0.01	0.31	0.30
Poland	Since 2000	Before 2000	Shorting allowed only in the permitted securities	41	17	0.04	0.01	0	0
Portugal	Since inception	None	Ban on naked short selling of specified financial stocks, September 24, 2008–present	1,384	1,082	1.40	1.24	0.03	0.03
Qatar ^a	Since inception	None		0	0	0	0	0	0
Russia	Pre-September 18, 2008; June 16, 2009–present	September 18, 2008–June 6, 2009	Uptick rule applies	73	50	0	0	1.47	0.82
Serbia ^a	None	Always		0	0	0	0	No ADRs	
Singapore	Since inception	None	Ban on naked short sales in buy-in market; onshore lending is limited while offshore lending is active	3,359	3,181	1.29	1.24	0.07	0.06
Slovakia ^a	None	Always		0	0	0	0	No ADRs	
Slovenia ^a	Since inception	None		0	0	0	0	No ADRs	
South Africa	Since inception	None	Naked short selling prohibited	2,259	1,744	0.33	0.32	1.02	0.93
South Korea	September 1996–September 30, 2008; June 1, 2009–present	Before 1996; October 1, 2008–May 31, 2009	Ban on shorting financial stocks, October 1, 2008–present; naked short selling, June 2000–present; uptick rule applies	3,633	2,987	0	0	0.76	0.54
Spain	Since 1992	Before 1992	Naked short selling prohibited	30,330	30,414	2.41	2.26	0.09	0.08
Sri Lanka ^a	None	Always		0	0	0	0	0	0
Sweden	Since 1991	Before 1991		11,065	9,467	2.63	2.08	1.39	0.91
Switzerland	Since inception	None	September 19, 2008–January 16, 2009, Swiss Federal Banking Commission and SIX Swiss Exchange prohibited naked short selling; SWX Europe also prohibited creation or increase of a net short position in certain specified UK and Swiss financial stocks	30,949	25,343	2.79	2.50	0.92	0.19
Taiwan	Pre-October 1, 2008; November 28, 2008–present	October 1, 2008–November 28, 2008	Uptick rule applies	301	195	0	0	0.76	0.59
Thailand	Since January 2001	Before January 2001	Only specified securities are eligible (underlying securities of SET 50 index, Exchange Traded Fund (ETF), and underlying securities of ETF); uptick rule applies; naked short selling prohibited	390	426	0.27	0.25	0	0
Tunisia ^a	None	Always		0	0	0	0	0	0
Turkey	Since inception	None	Uptick rule applies; only specified stocks eligible	481	457	0.28	0.25	0.38	0.26
Ukraine ^a	None	Always		0	0	0	0	0	0
UAE	None	Always		9	6	0.01	0	0	0
United Kingdom	Since inception	None	Ban on short selling of specified financial stocks, September 19, 2008–January 16, 2009	73,044	81,046	2.32	2.40	0.55	0.06
United States ^b	Since inception	None	Uptick rule effective, February 1, 1938–July 3, 2007; ban on naked short selling of 19 financial stocks, July 21, 2008–August 12, 2008; ban on short selling of specified financial stocks, September 19, 2008–October 8, 2008; quote based restrictions imposed, 2010	510,764	502,802	2.99	3.03	No ADRs	
Venezuela	None	Always		3	0	0	0	0	0
Zambia ^a	None	Always		0	0	0	0	No ADRs	
Zimbabwe ^a	None	Always		0	0	0	0	0	0

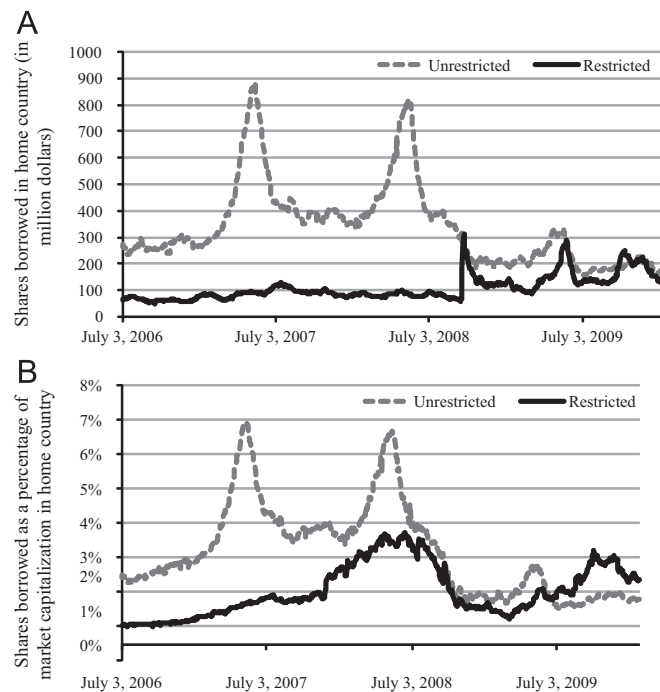


Fig. 1. Short selling in the home country, by short selling regime. Panel A plots short selling-related borrowing (amount outstanding at the end of each day, in millions of US dollars) for the countries in which short selling is restricted (solid line) and for the countries in which short selling is unrestricted (broken line). Short borrowing data are from Data Explorers for July 2006 to January 2010. Each observation in the original data set represents dollar stock borrowing for a given stock on a given day. Based on the originating country's short selling restriction on the relevant calendar day, each stock-day is allocated to either the restricted or the unrestricted portfolio. Then, for each country we compute the average stock borrowing across all stocks from that country and then the average across countries. Panel B scales the dollar borrowing by the stock's market capitalization.

presents specific regression equations assessing home market effectiveness of short selling restrictions and pairwise tests of worldwide regulatory reach. Appendix A shows our data sources and variable definitions.

4.1. Regulatory restrictions and other determinants of short selling

Our main dependent variable is short selling, which is measured with three alternative variables: stock borrowing in the home market, short interest of ADRs in the United States, and short volume of ADRs in the United States. Our key explanatory variable relates to the regulatory environment and captures information on whether short selling is legal or illegal and restricted or unrestricted.

The prior research suggests the need to include several control variables, although the nature of their relation with short selling is often unclear and few papers offer any direct test of how those variables affect short selling in a multi-market context. The presence of options trading is one such variable. Figlewski and Webb (1993) find a significantly higher average level of short interest for optionable stocks and argue that options facilitate short selling, which suggests a complementary effect. Similarly, Battalio and Schultz (2011) show complementarities between short selling and options. However, Bris, Goetzmann, and Zhu (2007) propose that options reduce short selling due to a substitution effect. Stock return

is another possible control variable, and Diether, Lee, and Werner (2009) as well as Jain, Jain, and McNish (2012) find that positive stock returns increase short selling. Blau, Van Ness, Van Ness, and Wood (2010), however, find that extremely negative market returns increase short selling. Diether, Lee, and Werner (2009) also show that return volatility affects short selling. Jones and Lamont (2002) find that stocks that are expensive to short or that enter the borrowing market have high valuations and low subsequent returns, consistent with the overpricing hypothesis. D'Avolio (2002) shows significant variations in borrowing costs across stocks, with a few stocks that become extremely special, demanding negative rebate rates (i.e., loan fees in excess of the risk-free rate). Krispy Kreme Doughnuts and Palm Inc. are examples of such stocks, exhibiting loan fees as high as 50% and 35%, respectively. To summarize, the evidence presented in these studies suggest that our model needs to include control variables for the existence of an option market in the home country, past returns, return volatility, and stock borrowing costs. We also include a number of other control variables that can relate to the level of short selling such as dividend yield, firm size (Diether, Lee, and Werner, 2009), the level of the ADRs, and days to cover.

D'Avolio (2002) and Asquith, Pathak, and Ritter (2005) argue that higher institutional ownership positively affects short selling by increasing the supply of loanable shares. Due to the limited availability of such data,

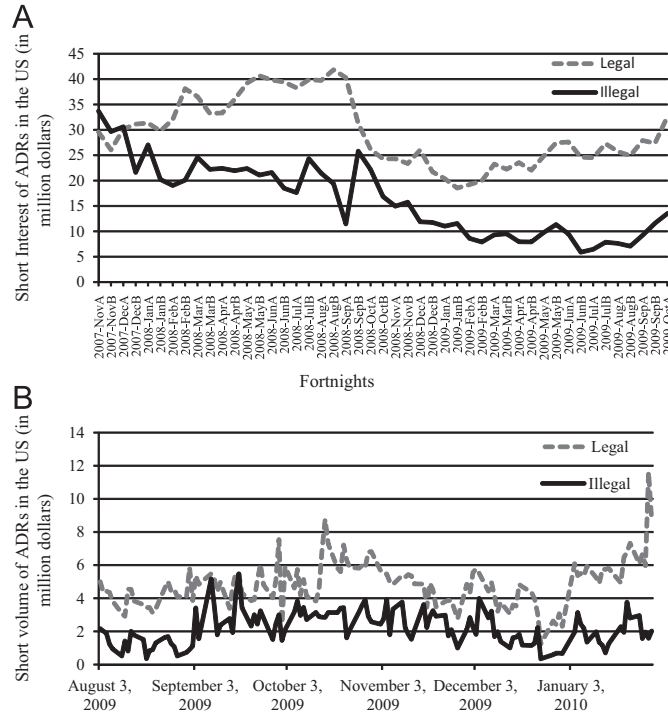


Fig. 2. Impact of home country regulations on monthly short selling of American Depository Receipts (ADRs) in the United States. Panel A plots the short interest of ADRs from countries in which short selling is illegal (solid line) and from countries in which short selling is legal (broken line). Short interest data are from shortsqueeze.com for November 2007 to October 2009. Panel B plots the short volume of ADRs from the countries in which short selling is illegal (solid line) and legal (broken line). Short volume data are from the Financial Industry Regulatory Authority and BATS Global Markets Exchanges for August 2009 to January 2010. Short interest data are at fortnightly frequency and short volume is at daily frequency. The computations follow a two-step averaging process analogous to that described in Fig. 1.

however, we use institutional ownership data only in a robustness test, and we expect a positive relation between institutional ownership and the amount of short selling.

4.2. Regulatory effectiveness of short selling restriction in curtailing home market stock borrowing

We assess the home market effectiveness of short selling restrictions using daily short selling-related borrowing data from Data Explorers. If the regulations are effectively enforced, countries with restrictions should have a lower amount of borrowing. Thus, we expect a statistically significant negative coefficient on *illegal* variable in the regression with stock borrowing as a dependent variable. We estimate all or a subset of the following equation and report the results in Table 2:

$$\begin{aligned} \text{scaled shares borrowed in home country}_{i,d} &= \alpha_0 + \alpha_1 \text{illegal}_{c,d} + \alpha_2 \text{uptick restriction}_{c,d} \\ &+ \alpha_3 \text{option markets}_{c,yr} + \alpha_4 \text{return}_{i,d} \\ &+ \alpha_5 \text{weekly return}_{i,d-1} + \alpha_6 \text{monthly return}_{i,d-1} \\ &+ \alpha_7 \text{half-yearly return}_{i,d-1} + \alpha_8 \text{scaled volume}_{i,d} \\ &+ \alpha_9 \text{dividend yield}_{i,d} + \alpha_{10} \text{return volatility}_{i,d} \\ &+ \alpha_{11} \text{borrowing cost}_{i,d} \\ &+ \alpha_{12-15} \text{industrial sector fixed effects}_i + \varepsilon_{i,d}, \end{aligned} \quad (1)$$

where α_0 – α_{15} are parameters to be estimated and ε is a random error term. Subscripts i , d , c , and yr denote firm, day,

country, and year, respectively. The subscript $d-1$ indicates that the return period ends on the day before day d . The dependent variable is *scaled shares borrowed in home country*, which is the home market borrowing of each individual firm on day t , scaled by its market capitalization. See Appendix A for other variable definitions and data sources.

Hereafter, standardized coefficient estimates are presented to allow comparison of the relative impact and importance of each determinant of short selling. These coefficients are estimates when all variables in the model are standardized to zero mean and unit variance, for the entire pooled data set, prior to performing the regression computations. Thus, the standardized intercept is 0.0000. The t -statistics for the standardized and unstandardized coefficients are the same. Furthermore, statistical significance is based on double clustered standard errors along firm and fortnightly time dimensions following Petersen (2009).

The coefficient on *illegal* is negative and significant in both Models 1 and 3 presented in Table 2. Similarly, the coefficient on the *uptick restriction* also is negative and significant in Models 2 and 3, indicating that short selling restrictions are associated with a lower level of stock borrowing in the home country. These results point to the regulatory effectiveness of short selling restrictions in curtailing borrowing activity in the home country.

Among the control variables, the coefficient on *option markets* is positive and significant in all models, indicating that derivative trades have the complementary effect of

Table 2

Regulatory effectiveness of home country short selling restrictions in curtailing home market stock borrowing.

For this table, our sample contains the underlying stocks for 1,601 American Depository Receipts for the period from July 2006 to January 2010. All our variables are defined in Appendix A. For each stock i for each day d , home market dollar stock borrowing is divided by stock i 's market capitalization to produce our dependent variable, the firm's scaled shares borrowed in the home country. Our main independent variables are the dummy variables illegal and uptick restriction. The remaining independent variables are option markets, return, weekly return, monthly return, half-yearly return, scaled volume, dividend yield, return volatility, borrowing cost, and sector fixed effects. Subscripts i , d , c , and yr denote firm, day, country, and year, respectively. All coefficients are standardized (standardized variables have zero mean and unit variance). Continuous variables are trimmed at the 1st and 99th percentiles. Statistical significance is based on double clustered standard errors along the firm and fortnightly time dimensions following Petersen (2009). ***significant at 0.01 level, **significant at 0.05 level, and *significant at 0.10 level.

Variable	Scaled shares borrowed in home country _{i,d}		
	Model 1	Model 2	Model 3
Intercept	0.0000	0.0000	0.0000
Illegal _{c,d}	−0.0063*** (−4.95)		−0.0017* (−1.69)
Uptick restriction _{c,d}		−0.0471*** (−5.22)	−0.0469*** (−5.20)
Option markets _{c,yr}	0.0321*** (7.15)	0.0156*** (3.27)	0.0156*** (3.29)
Return _{i,d}	−0.0067 (−1.38)	−0.0069 (−1.42)	−0.0069 (−1.42)
Weekly return _{$i,d-1$}	−0.0013 (−0.94)	−0.0011 (−0.90)	−0.0011 (−0.89)
Monthly return _{$i,d-1$}	−0.0089 (−1.23)	−0.0088 (−1.21)	−0.0088 (−1.22)
Half-yearly return _{$i,d-1$}	0.0110* (1.78)	0.0069 (1.20)	0.0067 (1.16)
Scaled volume _{i,d}	0.4589 (1.32)	0.4578 (1.31)	0.4578 (1.31)
Dividend yield _{i,d}	−0.0031 (−0.58)	−0.0132** (−2.32)	−0.0130** (−2.30)
Return volatility _{i,d}	−0.0031*** (−3.39)	−0.0019** (−2.46)	−0.0019** (−2.48)
Borrowing cost _{i,d}	0.0168 (1.38)	0.0138 (1.19)	0.0139 (1.20)
Days to cover _{$i,d-1$}	0.0114*** (4.41)	0.0095*** (4.35)	0.0095*** (4.35)
Sector fixed effects	Yes	Yes	Yes
Adjusted R-squared	0.2128	0.2146	0.2146
Number of observations	928,330	928,330	928,330

increasing stock borrowing in the home market. The coefficients of the return variables are not significant in any of the three models except for half-yearly return in Model 1. The insignificant coefficient on scaled volume indicates the lack of a strong relation between total volume and scaled stock borrowing, but our raw data suggest that the unscaled dollar borrowing is proportional to the stock's trading volume. Stock borrowing is lower for dividend payers and for volatile stocks. The coefficient on borrowing cost is not significant. The positive and significant coefficient on days to cover indicates that borrowers are not deterred by the length of time that it could take to cover the aggregate borrowed positions. Instead, our findings suggest that traders could apply momentum strategies and borrow previously borrowed stocks even more. Sector dummies for utilities,

transportation, insurance, and financial firms and the suppressed base case of industrials are included to capture industry fixed effects.

4.3. Impact of new restrictions on short selling in the home country and on ADRs

In this subsection, we test the regulatory reach hypothesis. We begin with univariate comparisons of differences in short selling-related borrowing in underlying markets and ADR markets during the most and least restrictive regimes, in turn.

Table 3 presents a pairwise comparison of the least restrictive and the most restrictive short selling regimes for the 18 countries that changed their regulations during our sample period.⁷ The different regulatory regimes in the order of increasing restrictiveness are no ban, uptick rule or naked ban, uptick and naked ban, and total ban on any type of short selling. Although regulatory effectiveness is not the main focus of our paper, our panel-data analysis of this issue is richer than previous work as prior studies focused on either a single country over time or a cross-section of data across multiple countries.

Short selling-related stock borrowing, scaled by market capitalization, is much higher in less restrictive regimes than in more restrictive regimes. This statement not only applies to the borrowing in underlying markets, but also extends to ADRs, consistent with our regulatory reach hypothesis. No evidence exists that traders try to move their activity internationally to avoid domestic restrictions. For example, when Australia restricted short selling, the ratio of shares borrowed to market capitalization dropped from 2.97% to 1.55%, representing a 48% decline. More important, the scaled short borrowing in Australian ADRs also declined from 0.070% to 0.035%, representing a 50% decline. The averages for the 18 countries in Table 3 support the regulatory reach hypothesis as well. When short selling is restricted, the short borrowing ratio declines by an average of 45% in the home markets and 68% in the ADR market. Both changes are statistically significant at the 0.01 level. Panel A uses underlying stock's market capitalization to scale the stock borrowing of both the ADRs and the underlying stock. Using this metric, the effects of regulations are larger in magnitude for home markets than for ADRs. Alternatively, because ADR volume is smaller than the volume in home markets, we also test the differences in stock borrowing scaled by ADR daily volume in Panel B.⁸ Average stock borrowing of ADRs drops from 6.64 times daily volume in the least restrictive regimes to 3.86 times daily volume in the most restrictive regimes, which represents a 42% decline.

⁷ Short selling restrictions also changed for China and Malaysia during our sample period, however, because we do not have stock borrowing data during different regulatory regimes for a pairwise comparison for these countries, we are unable to include these two countries in our pairwise comparisons.

⁸ Datastream, the Center for Research in Security Prices (CRSP), shortsqueeze.com, JP Morgan, and BNY Mellon do not report the ADR float. Because ADR volume is highly correlated with ADR float, however, we are able to proxy float data using volume data for ADRs and underlying stocks that are readily available from several data sources such as Datastream.

Table 3

Impact of home market short selling restrictions on stock borrowing in home country and American Depositary Receipt (ADR) borrowing in the United States.

Our sample for this table contains 1,695 underlying stocks and 853 ADRs from 18 countries that changed their short selling rules during July 2006 to January 2010. The acronym for restrictions (from the least restrictive to the most restrictive) in the nature of change column are as follows: no ban (Free), naked ban (N), uptick restriction (UR), naked ban plus uptick (N&UR), and total ban on any type of short selling (TB). The average stock borrowings from Data Explorer for all firms in a given country during the less (more) restrictive period are shown in Column 1 (2) for underlying shares and Column 5 (6) for ADRs. Column 3 (7) presents Column 2 minus 1 (6 minus 5). Column 4 (8) presents the values in Column 3 (7) converted to percentages. Panel A scales borrowing of both ADRs and underlying stock by market capitalization of the underlying stock. Panel B scales borrowing of underlying stock by daily trading volume of the underlying stock and borrowing of ADR by daily trading volume of ADR, each trimmed at 1st and 99th percentiles. ***Significant at 0.01 level, and **significant at 0.05 level.

Country	Nature of change	Underlying in home country				ADRs anywhere			
		Less restrictions (1)	More restrictions (2)	Difference Column 2 – 1 (3)	Percentage difference (4)	Less restrictions (5)	More restrictions (6)	Difference Column 6 – 5 (7)	Percentage difference (8)
Panel A: Scaled borrowing=dollar value of shares borrowed/market capitalization									
Australia	N-TB	2.979%	1.551%	– 1.427%	– 48%	0.070%	0.035%	– 0.035%	– 50%
Austria	Free-N	3.131%	1.521%	– 1.610%	– 51%	0.004%	0.001%	– 0.003%	– 70%
Belgium	Free-N	1.926%	0.716%	– 1.210%	– 63%	0.030%	0.000%	– 0.030%	– 100%
Denmark	Free-TB	2.607%	0.331%	– 2.277%	– 87%	0.057%	0.002%	– 0.056%	– 97%
France	Free-N	5.152%	4.494%	– 0.658%	– 13%	0.093%	0.005%	– 0.088%	– 95%
Germany	Free-N	4.981%	2.794%	– 2.187%	– 44%	0.070%	0.002%	– 0.068%	– 97%
Greece	N&UR-TB	0.103%	0.072%	– 0.031%	– 30%	0.038%	0.007%	– 0.031%	– 80%
Indonesia	Free-TB	0.043%	0.005%	– 0.038%	– 89%	0.252%	0.193%	– 0.059%	– 24%
Ireland	Free-TB	0.665%	0.079%	– 0.586%	– 88%	1.139%	0.018%	– 1.120%	– 98%
Italy	Free-N	3.300%	1.389%	– 1.910%	– 58%	0.031%	0.033%	0.002%	8%
Japan	UR-N&UR	1.458%	0.844%	– 0.614%	– 42%	0.020%	0.009%	– 0.012%	– 57%
Netherlands	Free-N	3.839%	2.560%	– 1.279%	– 33%	0.277%	0.070%	– 0.208%	– 75%
Norway	Free-N	3.361%	1.648%	– 1.713%	– 51%	0.028%	0.000%	– 0.028%	– 100%
Portugal	Free-N	1.368%	0.522%	– 0.846%	– 62%	0.028%	0.000%	– 0.028%	– 100%
South Korea	N&UR-TB	0.510%	0.203%	– 0.307%	– 60%	0.386%	0.252%	– 0.134%	– 35%
Switzerland	Free-N	3.578%	2.535%	– 1.044%	– 29%	0.108%	0.029%	– 0.079%	– 73%
Taiwan	N&UR-TB	0.073%	0.022%	– 0.052%	– 70%	0.362%	0.257%	– 0.105%	– 29%
United Kingdom	Free-TB	2.604%	1.694%	– 0.910%	– 35%	0.129%	0.080%	– 0.049%	– 38%
Weighted average		2.315%	1.277%	– 1.039%***	– 45%***	0.173%	0.055%	– 0.118%***	– 68%***
t Values–paired difference test				(– 6.15)	(– 10.43)			(– 1.97)	(– 8.70)
Panel B: Scaled borrowing=dollar value of shares borrowed/firm's daily trading volume									
Weighted average		42.65	15.01	– 27.64	– 65%***	6.64	3.85	– 2.79***	– 42%***
t Values–paired difference test				(– 1.09)	(– 3.19)			(– 3.40)	(– 3.24)

We also decompose the percentage decline in the stock borrowing in underlying and ADR markets into the five types of regulatory changes adopted by the countries listed in Table 3. For the first type of regulatory change, in which countries shift from no ban to a ban on naked shorts (Free-N), stock borrowing declines by 49% in the underlying market and 80% in the ADR market. For the second type of restriction, in which countries shift from a ban on naked shorts to a complete or total ban on any type of short selling (N-TB), stock borrowing declines by 48% in the underlying market and 50% in the ADR market. For the third type of restriction, in which countries shift from no ban to a complete ban (Free-TB), stock borrowing declines by 70% in the underlying market and 53% in the ADR market. For the fourth type of restriction, in which countries shift from a naked and uptick restriction to a complete ban (N&UR-TB), stock borrowing declines by 53% in the underlying market and 48% in the ADR market. Finally, when countries shift from uptick restriction to a naked and uptick restriction (UR-N&UR), stock borrowing declines by 42% in the underlying market and 57% in the ADR market. Each of the five types of increases in the

regulatory restrictions reduces short selling in both the underlying market and the ADR market simultaneously, consistent with the regulatory reach hypothesis.

4.4. Multivariate analysis of regulatory reach

In this subsection, we test the regulatory reach hypothesis in a multivariate setting by adding several firm-specific fundamentals or country characteristics. Both short interest and short volume data are analyzed. First, we estimate regressions based on all or a subset of the following equation using fortnightly short interest data from shortsqueeze.com:

$$\begin{aligned} \text{scaled ADR short interest}_{i,ft} = & \alpha_0 + \alpha_1 \text{illegal}_{c,ft} \\ & + \alpha_2 \text{uptick restriction}_{c,ft} + \alpha_3 \text{option markets}_{c,yr} \\ & + \alpha_4 \text{enforcement index}_c + \alpha_5 \text{restrictions}_{c,ft} \\ & \times \text{enforcement index}_c + \alpha_6 \text{fortnightly return}_{i,ft-1} \\ & + \alpha_7 \text{monthly return}_{i,ft} + \alpha_8 \text{half-yearly return}_{i,ft} \\ & + \alpha_9 \text{orthogonalized market capitalization}_i \end{aligned}$$

Table 4

Regulatory reach of home country short selling restrictions on American Depositary Receipts (ADR) short interest.

For this table, our sample contains 1,035 ADRs present in the triangular intersection of the shortsqueeze.com data set, Datastream International, and the initial ADR list for the period November 2007–October 2009. All of our variables are defined in Appendix A. Our dependent variable, Scaled ADR short interest, is the total short interest for firm i for fortnight ft divided by firm i 's number of shares outstanding. Our main independent variables are the dummy variables illegal and uptick restriction. The remaining independent variables are option markets, enforcement index, restriction $\times$ enforcement index, fortnightly return, monthly return, half-yearly return, orthogonalized market capitalization, scaled volume, dividend yield, level of ADR, return volatility, days to cover $_{i,ft-1}$, and sector fixed effects. Subscripts i , ft , c , and yr denote firm, fortnight, country, and year, respectively. All coefficients are standardized. Continuous variables are trimmed at the 1st and 99th percentiles. Statistical significance is based on double clustered standard errors along the firm and fortnightly time dimensions following Petersen (2009). ***Significant at 0.01 level, **Significant at 0.05 level, and *Significant at 0.10 level.

Variable	Scaled ADR short interest i_{ft}		
	Model 1	Model 2	Model 3
Intercept	0.0000	0.0000	0.0000
Illegal $_{c,ft}$	−0.0158** (−2.41)		−0.0141** (−2.05)
Uptick restriction $_{c,ft}$		−0.0198 (−1.42)	−0.0172 (−1.20)
Option markets $_{c,yr}$	0.0352* (1.66)	0.0288 (1.44)	0.0281 (1.41)
Enforcement index $_c$	0.0029 (0.12)	0.0036 (0.14)	0.0018 (0.07)
Restriction $_{c,ft} \times$ enforcement index $_c$	−0.0013 (−0.05)	0.0087 (0.37)	0.0083 (0.36)
Fortnightly return $_{i,ft-1}$	−0.0002 (−0.03)	−0.0001 (−0.01)	−0.0001 (−0.02)
Monthly return $_{i,ft}$	0.0079 (0.70)	0.0074 (0.66)	0.0076 (0.67)
Half-yearly return $_{i,ft}$	−0.0218 (−1.39)	−0.0209 (−1.33)	−0.0220 (−1.40)
Orthogonalized market capitalization $_i$	−0.0338*** (−3.28)	−0.0348*** (−3.38)	−0.0343*** (−3.33)
Scaled volume $_{i,ft}$	0.5535*** (9.09)	0.5535*** (9.10)	0.5534*** (9.10)
Dividend yield $_{i,ft}$	−0.0402*** (−2.96)	−0.0418*** (−3.04)	−0.0411*** (−2.99)
Level of ADR (1–3) $_i$	0.1047*** (3.98)	0.1048*** (3.99)	0.1050*** (4.00)
Return volatility $_{i,ft}$	−0.0246** (−2.31)	−0.0274** (−2.53)	−0.0257** (−2.36)
Days to cover $_{i,ft-1}$	0.1556* (1.81)	0.1554* (1.81)	0.1555* (1.81)
Sector fixed effects	Yes	Yes	Yes
Adjusted R-squared	0.3755	0.3754	0.3756
Number of observations	24,235	24,235	24,235

$$\begin{aligned}
& + \alpha_{10} \text{scaled volume}_{i,ft} + \alpha_{11} \text{dividend yield}_{i,ft} \\
& + \alpha_{12} \text{level of ADR}_i + \alpha_{13} \text{return volatility}_{i,ft} \\
& + \alpha_{14} \text{days to cover}_{i,ft} \\
& + \alpha_{15-18} \text{industrial sector fixed effects}_i + \varepsilon_{i,ft}, \quad (2)
\end{aligned}$$

where α_0 – α_{18} are parameters to be estimated and ε is a random error term. Subscripts i , ft , c , and yr denote firm, fortnight, country, and year, respectively. See Appendix A for variable definitions and data sources. To ensure that our regression results are robust to sector classification effects, we include sector dummies in all models, which are generated using information sourced from Datastream.

In Table 4, we present our regression results with scaled ADR short interest as the dependent variable. The coefficient on illegal is negative and significant in all specifications.⁹

⁹ The coefficient on illegal also is statistically significant at the 0.10 level or better if we use ordinary least squares standard errors, White's heteroskedasticity consistent errors, or errors clustered by firm and fortnight, by firm, by country and time, or by country.

Thus, if it is illegal to short sell a stock in its home country, short selling also is reduced for that country's ADRs in the United States, which is evidence in favor of the regulatory reach hypothesis.

The coefficient on uptick restriction, though negative in sign, is not statistically significant in these multivariate regressions. Because uptick restrictions still allow short sellers to trade, the impact of uptick restrictions is not as strong as a ban on short selling. The coefficient on option markets is positive and significant in Model 1, implying a complementary relation between shorting and derivatives, but the coefficient on option markets is not statistically significant in the remaining models.

Enforcement index is included as a control variable because stricter enforcement with less corruption incentivizes traders to follow and implement short selling rules more rigorously. We also use an interaction term between enforcement and short selling restrictions because enforcement by itself is not very important if there are no restrictions in the home market. We expect the interaction term to be negative, but it is statistically insignificant.

The coefficients of the return variables are not significant in any of the three models.¹⁰ The negative and significant coefficient on orthogonalized market capitalization indicates that scaled short interest is lower for bigger firms, but our raw data suggest that the unscaled short interest is higher for larger firms. The positive and significant coefficient on scaled volume indicates that scaled short interest is higher for firms with higher trading volume. The negative and significant coefficient on dividend yield indicates that short sellers do not maintain high open interest in high dividend paying firms.

There are three levels of ADRs, and level III ADRs have the most stringent requirement for trading in the United States. The level of ADR is included as a control variable to see if short sellers are trading level III ADRs more than level I or II ADRs. Its coefficient is positive and significant, indicating more short selling for higher level ADRs. The negative and significant coefficient on return volatility indicates that short sellers do not want to maintain high open interest for volatile stocks. The positive and significant coefficient on days to cover indicates that traders are not deterred by the length of time that it could take to cover the aggregate short positions outstanding in the ADRs. Instead, our findings demonstrate that traders apply momentum strategies and short previously shorted stocks even more.

We continue to test the regulatory reach hypothesis using high frequency short volume data from FINRA and the BATS Exchange. The transaction quantities for each stock for each day are aggregated to form stock-day observations for short selling volume. We estimate regressions based on all or a subset of the following equation:

$$\begin{aligned} \text{scaled ADR short volume}_{i,d} = & \alpha_0 + \alpha_1 \text{illegal}_{c,d} \\ & + \alpha_2 \text{uptickrestriction}_{c,d} + \alpha_3 \text{option markets}_{c,yr} \\ & + \alpha_4 \text{enforcement index}_{c,d} + \alpha_5 \text{restrictions}_{c,d} \\ & \times \text{enforcement index}_c + \alpha_6 \text{return}_{i,d} + \alpha_7 \text{weekly return}_{i,d-1} \\ & + \alpha_8 \text{monthly return}_{i,d-1} + \alpha_9 \text{half-yearly return}_{i,d-1} \\ & + \alpha_{10} \text{orthogonalized market capitalization}_i \\ & + \alpha_{11} \text{dividend yield}_{i,d} + \alpha_{12} \text{level of ADR}_i \\ & + \alpha_{13} \text{return volatility}_{i,d} + \alpha_{14-17} \text{industrial sector} \\ & \text{fixed effects}_i + \varepsilon_{i,d}, \end{aligned} \quad (3)$$

where α_0 – α_{17} are parameters to be estimated and ε is a random error term.

The estimation results for Eq. (3) are reported in Table 5. These data largely serve to confirm the evidence presented in Table 4. For brevity, we focus the discussion on Model 3, in which all variables are included.

The coefficient on illegal is negative and significant, indicating that higher levels of home country restrictions are associated with a lower level of short volume in the

ADR market. The coefficient on uptick restriction, though negative in sign, is not statistically significant again, implying that its impact is not as strong as a ban on short selling. The coefficient on option markets is insignificant. The coefficient on enforcement index itself is positive and statistically significant, but the key testable implication of enforcement is in the interaction variable, which again is insignificant. The positive and significant coefficients of shorter term return variables are consistent with prior studies that characterize short sellers as contrarian traders who short stock when it experiences a significant price appreciation (Diether, Lee, and Werner, 2009; Jain, Jain, and McNish, 2012). The effect of the monthly and half-yearly returns on short volume is insignificant. Coefficients of firm size, dividend yield, and ADR level are insignificant. Stock volatility has a positive coefficient in the volume regression, whereas it has a negative coefficient in the short interest regression. Thus, volatility induces short sellers to become active, but they are quick to cover their positions for such stocks, driving down the short open interest.

Our conclusions about the impact of home country regulations on ADRs using both fortnightly short interest data and high frequency FINRA short volume data are qualitatively similar, which is not surprising given the statistically significant positive correlation of 0.76 between these two alternative dependent variables. To estimate this correlation, we sum the FINRA short volume to calculate fortnightly short volume and then calculate the correlation between the two fortnightly variables. Overall, our results support the regulatory reach hypothesis.

4.5. Reverse regulatory reach

So far, our paper has analyzed the reach of home market restrictions on short selling of ADRs in the United States. However, the United States itself temporarily banned short selling of financial stocks on September 19, 2008. To test whether there is regulatory reach in the reverse direction, we investigate the effects of short selling restrictions in the United States on stock borrowing for US financial firms outside US markets.

Table 6 presents the results of an event study around the US short ban period, in which we analyze the stock borrowing for US stocks in domestic and foreign locations. We compute the changes in these variables from the period one month before to the period one month after September 19, 2008. The ban reduced scaled stock borrowing for the affected firms by 20% within the United States. No evidence emerges of any regulatory arbitrage in which traders moved their activity to foreign jurisdictions. Instead, the evidence points to reverse regulatory reach of US regulations in foreign locations, as stock borrowing for these US firms was reduced by 70% in foreign locations. Reverse regulatory reach applies to US banks as well as to nonbanking financial firms.

Additional insights into the issue of reverse regulatory reach for US financial stocks borrowed in foreign locations can be obtained by estimating the following multivariate

¹⁰ In un-tabulated analysis, we also consider the impact of home country restrictions on one month future return of ADRs. We find that future return of ADRs is negative when there are short sale restrictions in the home country. These findings are consistent with Asquith, Pathak, and Ritter (2005) who find that short sale constrained stocks underperform. Therefore, our results of regulatory reach are not driven by the lack of profitable opportunity for short sellers.

Table 5

Regulatory reach of home country short selling restrictions on American Depositary Receipts (ADR) short volume.

For this table, our sample contains 556 ADRs present in the Financial Industry Regulatory Authority (FINRA) and BATS Global Markets data sets for the period August 2009 to January 2010. All of our variables are defined in Appendix A. Our dependent variable, scaled ADR short volume, is daily short volume for each firm-day divided by firm i 's total trading volume. Our main independent variables are the dummy variables illegal and uptick restriction. The remaining independent variables are option markets, enforcement index, restriction $\times$ enforcement index, return, weekly return, monthly return, half-yearly return, orthogonalized market capitalization, dividend yield, level of ADR, return volatility, and sector fixed effects. Subscripts i , d , c , and yr denote firm, day, country, and year, respectively. All coefficients are standardized. Continuous variables are trimmed at the 1st and 99th percentiles. Statistical significance is based on double clustered standard errors along the firm and fortnightly time dimensions following Petersen (2009). ***-significant at 0.01 level, **-significant at 0.05 level, and *-significant at 0.10 level.

Variable	Scaled ADR short volume $_{i,d}$		
	Model 1	Model 2	Model 3
Intercept	0.0000	0.0000	0.0000
Illegal $_{c,d}$	-0.0292* (-1.74)		-0.0272** (-2.02)
Uptick restriction $_{c,d}$		-0.0264 (-0.45)	-0.0216 (-0.37)
Option markets $_{c,yr}$	-0.0686** (-2.30)	-0.0738 (-1.56)	-0.0769 (-1.60)
Enforcement index $_c$	0.1469** (2.34)	0.1477** (2.33)	0.1460** (2.31)
Restriction $_{c,d} \times$ enforcement index $_c$	0.0690 (1.42)	0.0831 (1.56)	0.0809 (1.54)
Return $_{i,d}$	0.0220** (2.12)	0.0222** (2.13)	0.0221** (2.11)
Weekly return $_{i,d-1}$	0.0382* (1.84)	0.0388* (1.88)	0.0385* (1.86)
Monthly return $_{i,d-1}$	-0.0543 (-0.87)	-0.0542 (-0.87)	-0.0542 (-0.87)
Half-yearly return $_{i,d-1}$	0.0196 (0.32)	0.0171 (0.27)	0.0184 (0.29)
Orthogonalized market capitalization $_i$	-0.0436 (0.00)	-0.0445 (0.00)	-0.0438 (0.00)
Dividend yield $_{i,d}$	-0.0146 (-0.72)	-0.0130 (-0.65)	-0.0143 (-0.71)
Level of ADR (1–3) $_i$	0.0101 (0.23)	0.0152 (0.36)	0.0125 (0.30)
Return volatility $_{i,d}$	0.0955*** (2.89)	0.0967*** (2.91)	0.0958*** (2.89)
Sector fixed effects	Yes	Yes	Yes
Adjusted R-squared	0.042	0.0416	0.0422
Number of observations	27,209	27,206	27,206

regression:

scaled shares borrowed in foreign locations $_{i,d}$

$$\begin{aligned}
 = & \alpha_0 + \alpha_1 \text{illegal}_d + \alpha_2 \text{option markets}_i \\
 & + \alpha_3 \text{return}_{i,d} + \alpha_4 \text{weekly return}_{i,d-1} \\
 & + \alpha_5 \text{monthly return}_{i,d-1} + \alpha_6 \text{half-yearly return}_{i,d-1} \\
 & + \alpha_7 \text{scaled volume}_{i,d} + \alpha_8 \text{dividend yield}_{i,d} \\
 & + \alpha_9 \text{return volatility}_{i,d} \\
 & + \alpha_{10} \text{stock borrowing cost}_{i,d} + \varepsilon_{i,d}, \quad (4)
 \end{aligned}$$

where α_0 – α_{10} are parameters to be estimated and ε is a random error term. Subscripts i and d denote firm and day, respectively. In this regression, the illegal dummy has a value of one during the period of the temporary ban on short selling in the United States from September 19, 2008 to October 8, 2008, and zero otherwise. The option market dummy has a value of one if the stock is optionable and zero otherwise. All variable definitions and data sources are in Appendix A.

The results are presented in Table 7 and reveal that the coefficient on illegal is negative and significant. This suggest that when short selling is temporarily banned in

the United States for a given firm, stock borrowing for that firm also is reduced in foreign locations. Thus, these results indicate that US regulations also have regulatory reach, which we call reverse regulatory reach.

The coefficient on option markets is positive and significant, implying a complementary relation between shorting and derivatives. The coefficients of the return variables are not statistically significant as are the coefficients of scaled volume, return volatility and borrowing cost variables. The negative and significant coefficient on dividend yield indicates that stock borrowing is lower for dividend payers.

Overall, our results reject the regulatory arbitrage hypothesis and support the regulatory reach hypothesis, including reverse regulatory reach. To provide further insights into these relations, we consider the mechanisms of regulatory reach.

4.6. Mechanisms of regulatory reach

The mechanisms of regulatory reach are grouped into two main categories: intergovernment cooperation and investor conduct. On the one hand, discussion groups

Table 6

Impact of Securities and Exchange Commission (SEC) temporary short selling ban on stock borrowing within and outside the United States.

We analyze scaled borrowing for the period 1 month before and 1 month after September 19, 2008, when the SEC imposed a temporary short selling ban for 799 financial firms including several American Depositary Receipts (ADRs). Of these, 539 firms have stock borrowing information in Data Explorers. Scaled borrowing (dollar value of shares borrowed/market capitalization) for the firms in the United States before the ban period (during the ban period) is reported in Column 1 (2). Scaled borrowing for the US firms outside the United States before the ban period (during the ban period) is reported in Column 5 (6). Column 3 (7) presents Column 2 minus 1 (6 minus 5). Column 4 (8) presents the values in Column 3 (7) converted to percentages. ***Significant at 0.01 level, and **Significant at 0.05 level.

	Domestic US stocks or ADRs in US				Domestic US stocks outside US or underlying in home country			
	Before ban	Ban	Column 3–2	Percentage difference	Before ban	Ban	Column 3–2	Percentage difference
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Overall	3.248%	2.590%	–0.657%*** (–13.77)	–20%	0.051%	0.015%	–0.036%* (–1.68)	–70%
<i>t</i> -Statistics								
Commercial banks	3.890%	3.210%	–0.680%	–17%	0.122%	0.027%	–0.095%	–78%
Saving Institutions and other banks	1.855%	1.446%	–0.410%	–22%	0.000%	0.001%	0.001%	0%
Banks	2.956%	2.400%	–0.556%	–19%	0.085%	0.019%	–0.066%	–78%
Insurance carriers	2.984%	2.227%	–0.757%	–25%	0.052%	0.016%	–0.036%	–69%
Other nonbanks	4.828%	3.773%	–1.055%	–22%	0.013%	0.011%	–0.002%	–17%
Nonbanks	4.011%	3.088%	–0.923%	–23%	0.030%	0.013%	–0.017%	–57%
ADRs	0.065%	0.040%	–0.025%	–38%	2.004%	1.282%	–0.723%	–36%

such as the G7, EU, and OECD, as well as bilateral investment treaties between countries, facilitate the implementation of regulatory intent through cross-border recognition of laws by enforcement agencies and courts. On the other hand, investors' own behavior is influenced by an adherence to ethical practices such as those mentioned in the CFA Institute's Code of Ethics and Standards Professional Conduct or conservative business practices driven by a desire to build an honest reputation. The analysis that follows discriminates between these two different reasons. Our proxy for the mechanism of intergovernment cooperation is the originating country's membership in G7, EU, or OECD and more formal bilateral investment treaties (BITs) between countries. For the investor conduct mechanism, our two proxies are concentration of CFA charterholders in the finance industry and the institutional infrastructure for stock borrowing. Neither of these two variables emanates directly from the short selling statutes, but both relate to the inherent desire of traders to do things in the right manner. We expect that CFA charterholders follow the professional conduct of knowing the laws in different jurisdictions and adhering to the stricter law in regard to short selling restrictions. Likewise, if institutions follow the spirit of the restrictions on short trades, they take the steps to shut down their stock lending operations even though the law might not prohibit lending itself.

This analysis of mechanism is based only on the observations that short selling is illegal in the home country (illegal=1) because we are interested in assessing the reach of regulatory restrictions, not the reach of regulatory freedom. Consequently, the illegal variable is not included in the regression model. The dependent variable is the scaled ADR short interest, and the independent variables capture

information on the possible mechanisms of regulatory reach as well as a number of control variables similar to those used previously:

$$\begin{aligned}
 \text{scaled ADR short interest}_{i,ft} = & \alpha_0 + \alpha_1 G7_c + \alpha_2 EU_c \\
 & + \alpha_3 OECD_c + \alpha_4 \text{group (G7/EU/OECD)}_c \\
 & + \alpha_5 \text{bilateral investment treaty with US}_c \\
 & + \alpha_6 \text{CFA/population}_c + \alpha_7 \text{option markets}_{c,yr} \\
 & + \alpha_8 \text{enforcement index}_c + \alpha_9 \text{fortnightly return}_{i,ft-1} \\
 & + \alpha_{10} \text{monthly return}_{i,ft} + \alpha_{11} \text{half-yearly return}_{i,ft} \\
 & + \alpha_{12} \text{orthogonalized market capitalization}_i \\
 & + \alpha_{13} \text{scaled volume}_{i,ft} + \alpha_{14} \text{dividend yield}_{i,ft} \\
 & + \alpha_{15} \text{level of ADR}_i + \alpha_{16} \text{return volatility}_{i,ft} \\
 & + \alpha_{17} \text{days to cover}_{i,ft} \\
 & + \alpha_{18-21} \text{industrial sector fixed effects}_i + \varepsilon_{i,ft},
 \end{aligned} \quad (5)$$

where α_0 – α_{21} are parameters to be estimated and ε is a random error term. Subscripts i , ft , c , and yr denote firm, fortnight, country, and year, respectively. See Appendix A for variable definitions and data sources.

Table 8 has the regression results for the mechanisms of regulatory reach. The first six models are regression equations that individually test our different measures of the mechanisms of regulatory reach. The coefficients for the measures of intergovernment cooperation are all negative and generally significant (only the coefficient on the bilateral investment treaty with the United States is insignificant in these stand-alone regressions). Further the CFA/population investor conduct variable is statistically significant and negative (–0.2074 in Model 6). Taken together, these results suggest that both intergovernment cooperation and investor conduct are effective

Table 7

Reverse regulatory reach of US short selling restriction outside United States.

For this table, our sample contains 168 domestic US firms that have borrowings outside the United States during July 2006–January 2010. All of our variables are defined in Appendix A. The dependent variable, scaled shares borrowed in foreign locations, is outstanding stock borrowing for each firm-day divided by each firm's market capitalization in the home market. Our main independent variable is the dummy variable illegal. The remaining independent variables are option, return, weekly return, monthly return, half-yearly return, scaled volume, dividend yield, return volatility, and borrowing cost. Subscripts i , d , c , and yr denote firm, day, country, and year, respectively. All coefficients are standardized. Continuous variables are trimmed at the 1st and 99th percentiles. Statistical significance is based on double clustered standard errors along the firm and fortnightly time dimensions following Petersen (2009). ***Significant at 0.01 level, **Significant at 0.05 level, and *Significant at 0.10 level.

Variable	Scaled shares borrowed in foreign locations i,d
Intercept	0.0000
Illegal	−0.0093** (−2.36)
Option $_i$	0.1623* (1.87)
Return $_{i,d}$	−0.0003 (−0.10)
Weekly return $_{i,d-1}$	−0.0007 (−0.08)
Monthly return $_{i,d-1}$	0.0068 (0.47)
Half-yearly return $_{i,d-1}$	0.0056 (0.12)
Scaled volume $_{i,d}$	0.0257 (1.04)
Dividend yield $_{i,d}$	−0.1178* (−1.65)
Return volatility $_{i,d}$	−0.0024 (−1.59)
Borrowing cost $_{i,d}$	0.0004 (0.03)
Adjusted R-squared	0.0466
Number of observations	13,229

mechanisms of regulatory reach. The final model (7) presented in Table 8 includes group (G7/EU/OECD), bilateral investment treaty with US, and CFA/population. Bilateral investment treaty is again negative in sign, and, in this model, it is statistically significant. The group variable again has a negative sign but is no longer significant.¹¹ The CFA/population investor conduct variable has a statistically significant negative coefficient in the combined regressions as well as in the stand-alone regression.

The proportion of CFA charterholders is a good proxy for investor conduct not only because of the CFA Institute's strong emphasis on high standards of ethics and professional conduct, but also because obtaining a CFA charter is a personal educational decision largely independent of the intergovernment cooperation among countries. Nonetheless, we also orthogonalize this investor conduct proxy to make it statistically independent of

the intergovernment cooperation in a two-step regression. For the sake of brevity, these results are not tabulated. However, the coefficient on the orthogonalized CFA variable is significantly negative, indicating that investor conduct is a driving force behind the regulatory reach of foreign laws.

Legal restrictions in the home market limit the shares institutions make available for borrowing, aiding regulatory reach. We test and confirm this in the data in which a regression of scaled shares available for lending on the illegal variable produces a negative coefficient.¹² Thus, when a country bans short selling, fewer shares of that country's stocks are available worldwide for short sellers to borrow, which again results in short sellers being less aggressive.

4.7. Economic significance of the results

While the layers of scaling, standardization, and clustering presented in our regressions tables are an econometric necessity, they can make our results difficult to interpret. We address this issue by discussing the economic significance of our results.

The unconditional mean of short interest plotted in Fig. 2, Panel A is \$30 million per stock for ADRs from countries in which it is legal to short sell and \$16 million for ADRs from countries in which it is illegal. Thus, the unconditional economic significance of home country short selling restriction is a reduction in ADR short selling by \$14 million. In terms of our regression models, the economic significance of regulatory reach can be reverse engineered from the standardized coefficients reported in Panel C of Table 4. To compare the predicted value of dollar short interest when illegal equals one with its value when illegal equals zero, these standardized values must be transformed to nominal (unstandardized) values. This transformation can be achieved in the following way. First, calculate the predicted value of standardized scaled short interest by multiplying the standardized coefficient on illegal (−0.0141) with the standardized value of illegal when illegal equals zero; i.e., the nominal value (0) minus the mean of illegal (0.0728) divided by the standard deviation of illegal (0.2598). Second, transform standardized scaled short interest to an unstandardized scaled short interest ratio by multiplying it with the standard deviation of scaled short interest (0.0149) and adding the mean of scaled short interest (0.0028). Third, repeat these two steps to obtain the unstandardized scaled short interest ratio when illegal equals one. Fourth, calculate

¹¹ We also use executive membership of International Organization of Securities Commissions as an additional proxy for intergovernment cooperation. The coefficient on the IOSCO dummy variable is insignificant both with and without any control variables.

¹² The results are not tabulated for brevity but are summarized in the following equation. Scaled shares available to borrow worldwide $_{i,d}$ =0.0000 (intercept)−**0.0811** illegal $_{c,d}$ +**0.6790** uptick restriction $_{c,d}$ +**0.0259** option markets $_{c,yr}$ −0.0012 return $_{i,d}$ +**0.0156** weekly return $_{i,d}$ −0.0087 monthly return $_{i,d}$ +0.0194 half yearly return $_{i,d}$ +**0.0858** scaled volume $_{i,d}$ −**0.0476** dividend yield $_{i,d}$ −0.0339 return volatility $_{i,d}$ −**0.0422** borrowing cost $_{i,d}$ −0.0035 days to cover $_{i,d}$ −1+sector fixed effects+ $\varepsilon_{i,d}$, where bold indicates significance at the 0.10 level or better using double cluster standard errors. *Shares available to borrow worldwide* is the value of current inventory available from beneficial owners for loans scaled by market capitalization of the firm.

Table 8

Mechanism of regulatory reach of home country short selling restrictions on American Depositary Receipt (ADR) short interest.

For this table, our sample contains 196 ADRs for which illegal equals one, which is a subset of the 1,035 ADRs present in the triangular intersection of the shortsqueeze.com data set, Datastream International, and the initial ADR list from November 2007 to October 2009. Our dependent variable, scaled ADR short interest, is total short interest for each firm-fortnight divided by each firm's shares outstanding. Our main independent variables are G7, European Union (EU), Organization for Economic Co-operation and Development (OECD), group (G7/EU/OECD), bilateral investment treaty with United States, and CFA/population. The remaining independent variables are option markets, enforcement index, fortnightly return, monthly return, and half-yearly returns, orthogonalized market capitalization, scaled volume, dividend yield, level of ADR, return volatility, days to cover_(t-1), and sector fixed effects. Subscripts *i*, *ft*, *c*, and *yr* denote firm, fortnight, country, and year, respectively. All coefficients are standardized. Continuous variables are trimmed at the 1st and 99th percentiles. Statistical significance is based on double clustered standard errors along the firm and fortnightly time dimensions following Petersen (2009). ***Significant at 0.01 level, **Significant at 0.05 level, and *Significant at 0.10 level.

Variable	Scaled ADR short interest _{<i>i,ft</i>}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Intercept	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
G7 _{<i>c</i>}	-0.0518** (-2.41)						
EU _{<i>c</i>}		-0.0865** (-2.58)					
OECD _{<i>c</i>}			-0.1352* (-1.81)				
Group (G7/EU/OECD) _{<i>c</i>}				-0.1372* (-1.83)			-0.1226 (-1.37)
Bilateral investment treaty with US _{<i>c</i>}					-0.0672 (-1.38)		-0.1068** (-1.97)
CFA/population _{<i>c</i>}						-0.2074*** (-2.80)	-0.0807* (-1.93)
Option markets _{<i>c,yr</i>}							0.1334 (1.16)
Enforcement index _{<i>c</i>}							-0.0451 (-0.47)
Fortnightly return _{<i>i,ft-1</i>}							0.0070 (0.52)
Monthly return _{<i>i,ft</i>}							-0.0028 (-0.15)
Half-yearly return _{<i>i,ft</i>}							-0.0015 (-0.07)
Orthogonalized market capitalization _{<i>i</i>}							-0.0096 (-0.22)
Scaled volume _{<i>i,ft</i>}							0.7712*** (4.05)
Dividend yield _{<i>i,ft</i>}							0.0033 (0.14)
Level of ADR (1–3) _{<i>i</i>}							0.0054 (0.10)
Return volatility _{<i>i,ft</i>}							-0.0356* (-1.71)
Days to cover _{<i>i,ft-1</i>}							0.0936*** (2.66)
Sector fixed effects							Yes
Adjusted R-squared	0.0023	0.0071	0.0179	0.0184	0.0041	0.0426	0.6844
Number of observations	2,389	2,389	2,389	2,389	2,389	2,389	1,197

the difference between unstandardized scaled short interest ratio when illegal equals zero and unstandardized scaled short interest ratio when illegal equals one. Finally, convert unstandardized scaled short interest ratio to dollar short interest by simultaneously multiplying it by mean of shares outstanding (1,027 million) and mean of price (\$24.11) to obtain the economic significance of regulatory reach which is \$20 million reduction in an average ADR's short interest in this example (a technical Internet Appendix A is available for our paper on the journal website <http://jfe.rochester.edu/data.htm>, which contains additional data for figures and a spreadsheet explaining in greater detail the computations for economic significance). Similarly, transforming the

standardized coefficient on -0.0017 for illegal variable in Table 2, Model 3, implies that home country short selling restrictions reduce outstanding shares borrowed by \$14 million per stock. Transforming the standardized coefficient on -0.0272 for the illegal variable in Table 5, Model 3, implies that home country short selling restrictions reduce ADR short volume by \$1.9 million per stock per day. Transforming the standardized coefficient on -0.0093 for the illegal variable in Table 7 implies that short selling ban in the United States reduced outstanding shares borrowed for domestic US firms in foreign locations by \$0.74 million per stock. Transforming the standardized coefficient on -0.0807 for CFA/population in Table 8 implies that when a country's proportion

of charterholders doubles, short interest of ADRs from that country reduces by \$4 million per stock, if is illegal to short sell in the home market.

4.8. Robustness tests

The negative coefficient on illegal in the regulatory reach regression of scaled ADR short interest is robust to alternative model specifications and subsamples. For the sake of brevity, these results are not formally tabulated, but instead our findings are summarized in this subsection. In the discussion of robustness tests, the benchmark value for the coefficient on illegal is -0.0141 from Model 3 of Table 4.

First, we estimate the regression with unscaled ADR short interest without adjusting for the number of shares outstanding. The coefficient on illegal is -0.0363 , which suggests a slightly stronger regulatory reach on unscaled ADR short interest than on scaled ADR short interest. Second, we estimate the scaled short interest regression using a smaller sample of 5,974 observations for only level III ADRs, which are subject to the most stringent US securities regulations and trade alongside other US stocks on the main stock exchanges. The coefficient on illegal in this stand alone regression is -0.0628 , implying that even level III ADRs are affected by home country short selling regulations. Third, we control for institutional ownership, which has a positive relation with short selling, consistent with prior research. Due to limited data availability, when institutional ownership is added as the only control variable in the regression, the sample size still drops dramatically from 24,235 in Table 4 to 9,758. This does not change the tenor of our results, however, as the coefficient on illegal remains negative (-0.0427) and statistically significant. The coefficient on institutional ownership itself is 0.2352 and statistically significant. The positive coefficient is consistent with D'Avolio (2002) and Asquith, Pathak, and Ritter (2005), who argue that higher institutional ownership positively affects short selling by increasing the supply of loanable shares.

We also estimate regulatory reach regression with 45 country level and 1,938 country-fortnight observations instead of 24,235 stock fortnight regression observations used in Table 4. For these equal-weighted (by country) regressions, all variables are averaged by country or by country-fortnight. The coefficients of illegal are -0.2530 and -0.1098 , and they are statistically significant at 0.01 level. White's heteroskedasticity consistent standard errors are used for these country-level regressions as there are no firm clusters. These country-level results indicate regulatory reach even more strongly.

Next, we test the regulatory reach hypothesis for the three subperiods before, during, and after the 2008 financial crisis. We estimate these three models with unscaled short interest as the dependent variables. The estimated coefficients of the illegal variable are negative and statistically significant in all three regressions ranging from -0.0672 to -0.0227 , compared with the overall period coefficient on -0.0363 . Similarly, home country short selling restrictions have a negative impact on scaled short interest, both including and excluding the crisis

period. The results indicate that restrictions curtail short selling in the ADR markets and that the financial crisis did not materially change the phenomenon of regulatory reach.

Finally, we perform a control sample analysis to ensure that the negative coefficient on illegal is based on regulatory reach of home country restrictions and not on any confounding factors within the United States that would affect all domestic stocks as well as the ADRs. Each ADR is matched with a non-ADR domestic US stock. Our matching criteria include industry, exchange listing, price-to-book value (PTBV), and market capitalization.¹³ Short selling of US stocks is not expected to be affected by changes in the legality of short selling abroad for the matched stocks. The period when short selling was restricted in the United States is excluded. ADR stocks are similar to the matched control sample of US stocks except that some ADRs originate from a home country with short selling restrictions and others originate from unrestricted countries. Using this control sample, we estimate a regression with scaled short interest as the dependent variable.¹⁴ The illegal variable in this model is interacted with the indicator variables for ADRs (treatment stocks) or domestic stocks (matched control sample). The home country regulations reach out in lowering the ADR short interest (with a negative coefficient on -0.0347 that is statistically significant at the 0.01 level). However, those regulations appear to have no impact on short interest of matched non-ADR domestic US stocks (the coefficient 0.0190 is not significant even at the 0.10 level).

Thus, the additional analysis discussed in this subsection indicates that our results on regulatory reach are robust for different specifications and subsamples.

5. Summary and conclusions

Our paper provides the first comprehensive characterization of the worldwide market for short selling and stock borrowing using several data sets that capture short selling regulations, actual short selling, and outstanding short interest in stocks from 82 countries. The theme of our paper is the examination of global regulatory reach of short selling restrictions in a multi-market environment.

The first issue that we address is whether restrictions on short selling in a given country are effective in reducing short selling-related borrowing. Our results

¹³ For the price to book value for ADRs (non-ADR domestic US firms) the mean is 3.13 (3.13) and the standard deviation is 6.81 (5.91). For the natural log of market value for ADRs (non-ADR domestic US firms), the mean is 7.80 (7.35) and the standard deviation is 1.99 (1.79).

¹⁴ The results are not tabulated for brevity, but the regression estimates can be summarized as follows. Scaled short interest_{it} = 0.0000 intercept -0.0347 illegal_{c,ft} $\times$ ADR + 0.0190 illegal_{c,ft} $\times$ non-ADR domestic US stocks -0.0153 uptick restrictions_{c,ft} -0.0558 option market-_{sc,yr} + **0.0577** enforcement index_c + 0.0188 restriction_{c,d} $\times$ enforcement index_c + 0.0002 fortnightly return_{it-1} -0.0052 monthly return_{it} **-0.0236** half yearly return_{it} + **0.0043** orthogonalized market capitalization_{it} + **0.3808** scaled volume_{it} -0.0307 dividend yield_{it} -0.0298 return volatility_{it} + **0.2931** days to cover_{it-1} + industry fixed effects + ε_{it} , where bold indicates significance at the 0.10 level or better.

Table A1

Variable	Definition	Source
Measures of short selling (dependent variables)		
Shares borrowed	Total value of borrowed or loaned securities net of double counting, reported at daily frequency	Data explorers
Total short interest	Total number of outstanding shorted shares for each American Depository Receipt (ADR) reported at fortnightly frequency	shortsqueeze.com
Short volume	Aggregate number of shares sold short each day (computed from transaction level data)	Regulation SHO data from Financial Industry Regulatory Authority and BATS Global Markets exchange
Regulatory measures		
Illegal	Indicator variable that has value of one when short selling is banned in the home country and zero when short selling is allowed, based on academic articles and practitioners reports such as 2009 client briefing by Clifford Chance LLP http://www.cliffordchance.com/publicationviews/publications/2009/07/clifford_chance_commentsshortsellingrules.html	Bris, Goetzmann, and Zhu (2007), Charoenrook and Daouk (2009), 2009 client briefings by Clifford Chance LLP, Beber and Pagano (2013), and direct correspondence with stock exchanges and regulators
Uptick restriction	Indicator variable that has value of one when short selling is restricted by price tests in the home country and zero otherwise	Client briefing by Clifford Chance LLP and stock exchange websites
Restriction	Indicator variable that has value of one when short selling is restricted by any means such as total ban, ban on naked short selling, or price tests in the home country and zero otherwise	Same sources as illegal variable
Country and market design variables		
Option markets	Number of years for which the option markets have existed in a particular country	Charoenrook and Daouk (2009)
Option	Equals one if traded on the Chicago Board of Exchange (CBOE), and zero otherwise	CBOE option trading list
Enforcement index (CPI)	Country score of perceived level of corruption relating to bribery of public officials, kickbacks in public procurement, embezzlement of public funds, and effectiveness of public sector anticorruption efforts. Stronger enforcement > corruption and lawlessness	Transparency International http://www.transparency.org/policy_research/surveys_indices/cpi
G7	Indicator variable (one for G7 countries)	
EU	Indicator variable (one for European Union countries)	
OECD	Indicator variable (one for Organization for Economic Co-operation and Development countries)	
Group (G7/EU/OECD)	Indicator variable (one for G7, EU, or OECD group countries)	
Bilateral investment treaty with US	Indicator variable (one when the country has signed a bilateral investment treaty with the United States)	United Nations Conference on Trade and Development
CFA/population	Number of charterholders in the country divided by the population of that country	CFA Institute member resources and www.worldatlas.com
Firm-specific variables		
Return variables		
Return (daily or fortnightly)	$\text{Log}(\text{Return index}_t) - \text{Log}(\text{Return index}_{t-1})$. For daily returns, substitute d for t . For fortnightly returns, substitute ft for t . The Return index is adjusted for dividends. These are essentially a firm's stock returns	Datastream International
Weekly return	$\text{Log}(\text{Return index}_{d-1}) - \text{Log}(\text{Return index}_{d-6})$	Datastream International
Monthly return	$\text{Log}(\text{Return index}_{d-1}) - \text{Log}(\text{Return index}_{d-21})$	Datastream International
Half-yearly return	$\text{Log}(\text{Return index}_{d-1}) - \text{Log}(\text{Return index}_{d-121})$	Datastream International
Orthogonalized market capitalization	Share price multiplied by the number of ordinary shares outstanding in millions of US dollars at the beginning of our sample period. For Table 4, market capitalization is orthogonalized by regressing it on shares outstanding and trading volume. For Table 5, market capitalization is orthogonalized by regressing it on trading volume	Datastream International
Scaled volume	For daily-level borrowing data regressions, dollar value of trading volume on day t divided by market capitalization. For fortnightly short interest regressions, trading volume in terms of number of shares on the last day of fortnight divided by shares outstanding	Datastream International
Dividend yield	Dividend per share as a percentage of the firm's share price	Datastream International
Level of ADR	Ordinal variable that equals 1(OTC), 2 (exchange listed), or 3 (public offering to raise capital)	www.adrbnymellon.com (BNY Mellon), www.adr.com (JP Morgan)
Return volatility	For daily regressions, intraday high price minus intraday low price divided by intraday high price. For fortnightly regressions, fortnightly high price minus fortnightly low price divided by fortnightly high price	Datastream International
Borrowing costs		Data Explorers

Table A1 (continued)

Variable	Definition	Source
	Value-weighted average stock lending fee for all open loans expressed in undisclosed fee buckets 0–5 (0 indicates the cheapest to borrow)	
Lagged days to cover (DTC)	Current short interest divided by average daily trading volume (ratio computed for previous day)	shortsqueeze.com
Sector fixed effects	Dummy variables for utility, transportation, financial, and insurance sectors. Base case is the industrials sector	Datastream International

suggest that short selling regulations reduce borrowing in the home country by 45%, on average.

Next, we examine whether home country restrictions on short selling reduce short selling of ADRs in the United States, which we call the regulatory reach hypothesis. Alternatively, the competing hypothesis of regulatory arbitrage takes the view that if short selling is illegal in a home country, there is greater short selling in the United States of the ADRs from that country due to short sellers' efforts to circumvent the home country regulations. Using data from shortsqueeze.com, FINRA, BATS, and Data Explorers, we find support for the regulatory reach hypothesis in cross-sectional, event study, and panel-data tests. Due to the regulatory reach of home country short selling restrictions, stock borrowing for the affected ADRs drops by 68%, on average, in the United States. Similarly, in a multivariate regression of scaled short interest, the statistically significant negative coefficient again supports our regulatory reach hypothesis. Our results are robust to differences in firm-specific characteristics, such as option market availability, past returns, firm size, trading volume, dividend yield, return volatility, days to cover, level of ADR, and industrial sectors. We also show a reverse regulatory reach for the SEC's temporary short selling ban on shorting 799 financial stocks in 2008, which reduced stock borrowing for these firms both within and outside the United States.

Various mechanisms can cause home market regulations to have a strong global regulatory reach, and we consider two possible mechanisms: intergovernment cooperation and investor conduct. Intergovernment cooperation can work through working groups such as G7, OECD, or EU or bilateral investment treaties. Investor conduct is influenced by professional standards of bodies such as the CFA Institute or institutional unwillingness to conduct stock lending when short selling is illegal. Our results indicate that investor and institutional conduct strengthens regulatory reach more than membership in working groups or treaties.

Our paper contributes to the short selling and cross-listing research with two main findings. First, restrictions on short selling in a given country are effective in reducing short selling-related borrowing in that country. Second, from the investors' point of view, cross-listing in the United States is not a vehicle for circumventing regulatory control on short selling in the home country. Regulatory controls in the home country also stifle short selling of ADRs. An important implication of our regulatory reach findings is that the regulators appear to have

a great ability to obtain desired investor protection outcomes with little sacrifice on the business competitiveness front even in a fiercely competitive global economy. Additional cooperation and coordination among the global enforcement agencies can further strengthen regulatory reach.

Appendix A

A.1. Variables definitions and data sources

See Table A1.

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